

Carathéodory Extension Theorem

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1 Outer Measure

Definition 1.1. (Outer measure.) Let Ω be a set. Then, $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ is called an *outer measure* on Ω if:

1. $\mu^*(\emptyset) = 0$.
2. (*Monotonicity.*) If $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$.
3. (*Sub-additivity.*) If $E \subseteq \bigcup_{j \geq 1} E_j$, then $\mu^*(E) \leq \sum_{j \geq 1} \mu^*(E_j)$.

Proposition 1.2. Let $\mathcal{C}$ be a collection of subsets of Ω such that $\emptyset, \Omega \in \mathcal{C}$. Let $\rho : \mathcal{C} \rightarrow [0, \infty]$ such that $\rho(\emptyset) = 0$. Define $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ by

$$\mu^*(A) := \inf \left\{ \sum_{j \geq 1} \rho(A_j) : A \subseteq \bigcup_{j \geq 1} A_j \text{ and } A_j \in \mathcal{C} \forall j \geq 1 \right\}.$$

Then, μ^* is an outer measure on Ω .

Proof. We first note that since $\Omega \in \mathcal{C}$, for any $A \in \mathcal{P}(\Omega)$, $A \subseteq \Omega$ and so we can set $A_1 = \Omega$ and $A_j = \emptyset$ for all $j \geq 2$. Then, $A \subseteq \bigcup_{j \geq 1} A_j$ and hence μ^* is well-defined.

We show that it is an outer measure as follows:

1. We start by showing that $\mu^*(\emptyset) = 0$. Set $A_j = \emptyset$ for all $j \geq 1$. Then, $\emptyset \subseteq \bigcup_{j \geq 1} A_j$ and so $\mu^*(\emptyset) \leq \sum_{j \geq 1} \rho(A_j) = 0$ since $\rho(\emptyset) = 0$. Further, if $\{A_j\}_{j \geq 1}$ is any covering of $\emptyset$, then $\sum_{j \geq 1} \rho(A_j) \geq 0 \implies \mu^*(\emptyset) \geq 0$. Hence, $\mu^*(\emptyset) = 0$.
2. We now show monotonicity: let $A \subseteq B$. We want to show that $\mu^*(A) \leq \mu^*(B)$. Let $\{B_j\}_{j \geq 1}$ be a covering of B . Then, $B \subseteq \bigcup_{j \geq 1} B_j$, but since $A \subseteq B$, we have that $A \subseteq \bigcup_{j \geq 1} B_j$, that is, $\{B_j\}_{j \geq 1}$ is also a covering of A . Thus, every covering of B is a covering of A , which implies that $\mu^*(A)$ is the infimum taken over a set which contains the set over which the infimum is taken for $\mu^*(B)$. Hence, $\mu^*(A) \leq \mu^*(B)$.

3. Finally, we show sub-additivity: if $A \subseteq \bigcup_{j \geq 1} A_j$, then $\mu^*(A) \leq \sum_{j \geq 1} \mu^*(A_j)$. We may assume that $\mu^*(A_j) < \infty$ for all $j \geq 1$, for if that is not the case then the assertion is trivial.

By definition of μ^* , for all $\varepsilon > 0$, there exists some covering $\{E_{j,k}\}_{k \geq 1}$ of A_j in $\mathcal{C}$ such that

$$\mu^*(A_j) \leq \sum_{k \geq 1} \rho(E_{j,k}) \leq \mu^*(A_j) + \frac{\varepsilon}{2^j}.$$

Now, $A \subseteq \bigcup_{j \geq 1} A_j \subseteq \bigcup_{j,k \geq 1} E_{j,k}$, and therefore,

$$\mu^*(A) \leq \sum_{j,k \geq 1} \rho(E_{j,k}) \leq \sum_{j \geq 1} \left(\mu^*(A_j) + \frac{\varepsilon}{2^j} \right) = \sum_{j \geq 1} \mu^*(A_j) + \varepsilon.$$

Since this is true for all $\varepsilon > 0$, we obtain $\mu^*(A) \leq \sum_{j \geq 1} \mu^*(A_j)$, as required.

Hence, μ^* is an outer measure on Ω . □

2 μ^* -measurable

Definition 2.1. Let μ^* be an outer measure on Ω . Then, $A \subseteq \Omega$ is said to be μ^* -measurable or *Carathéodory-measurable* if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for all $E \subseteq \Omega$.

Theorem 2.2. (Carathéodory's Extension Theorem.) Let μ^* be an outer measure on Ω and let $\mathcal{M}$ be the collection of all μ^* -measurable sets. Then:

1. $\mathcal{M}$ is a σ -algebra.
2. The restriction of μ^* on $\mathcal{M}$, $\mu^*|_{\mathcal{M}}$, is a complete measure.

Proof. 1. We first show that $\mathcal{M}$ is a σ -algebra.

Note that for any $E \subseteq \Omega$, $E \subseteq (E \cap A) \cup (E \cap A^c)$ and therefore $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ due to the sub-additivity of μ^* . Hence, to show that $A \in \mathcal{M}$, it suffices to show that $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. Since this is trivial if $\mu^*(E) = \infty$, we consider the case where $\mu^*(E) < \infty$.

- (a) Since $\mu^*(E \cap \emptyset) + \mu^*(E \cap \emptyset^c) = \mu^*(\emptyset) + \mu^*(E) = 0 + \mu^*(E) = \mu^*(E)$ for all $E \subseteq \Omega$, $\emptyset \in \mathcal{M}$.

- (b) $\mathcal{M}$ is closed under complements since the definition of μ^* -measurable sets is symmetric about complements: if $A \in \mathcal{M}$, then for all $E \subseteq \Omega$,

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A^c) + \mu^*(E \cap A) \\ &= \mu^*(E \cap A^c) + \mu^*(E \cap (A^c)^c)\end{aligned}$$

and therefore $A^c \in \mathcal{M}$.

- (c) It remains to show that $\mathcal{M}$ is closed under countable unions. If $A_j \in \mathcal{M}$ for all $j \geq 1$, we want to show that $\bigcup_{j \geq 1} A_j \in \mathcal{M}$.

In order to show this, we first prove closure under finite unions. Let $A, B \in \mathcal{M}$. We want to show that $A \cup B \in \mathcal{M}$, that is,

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

for all $E \subseteq \Omega$. Since $A \in \mathcal{M}$,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Further, since $B \in \mathcal{M}$,

$$\mu^*(E \cap A^c) \geq \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c)$$

and so,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).$$

Equivalently,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap (A \cup B)^c). \quad (1)$$

But since $A \cup B = A \cup (B \setminus A) = A \cup (B \cap A^c)$,

$$E \cap (A \cup B) = E \cap (A \cup (B \cap A^c)) = (E \cap A) \cup (E \cap A^c \cap B).$$

Due to the sub-additivity of μ^* ,

$$\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B).$$

Using this in (1) yields $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Since this is true for all $E \subseteq \Omega$, $A \cup B \in \mathcal{M}$. We have thus shown closure under finite unions, implying that $\mathcal{M}$ is an algebra.

We now show closure under countable unions. Let $A = \bigcup_{j \geq 1} A_j$. We want to show:

$$\mu^*(E) \geq \mu^*\left(E \cap \bigcup_{j \geq 1} A_j\right) + \mu^*\left(E \setminus \bigcup_{j \geq 1} A_j\right).$$

Due to closure under finite unions, we know:

$$\mu^*(E) \geq \mu^*\left(E \cap \bigcup_{j=1}^n A_j\right) + \mu^*\left(E \setminus \bigcup_{j=1}^n A_j\right).$$

Since $\bigcup_{j=1}^n A_j \subseteq \bigcup_{j \geq 1} A_j$, $\mu^*(E \setminus \bigcup_{j=1}^n A_j) \geq \mu^*(E \setminus \bigcup_{j \geq 1} A_j)$.
Hence,

$$\mu^*(E) \geq \mu^*\left(E \cap \bigcup_{j=1}^n A_j\right) + \mu^*\left(E \setminus \bigcup_{j \geq 1} A_j\right). \quad (2)$$

Define $B_1 = A_1$, $B_2 = A_2 \setminus A_1$, and similarly, $B_j = A_j \setminus \bigcup_{k=1}^{j-1} A_k$. Clearly, B_j are pairwise disjoint and $B_j \in \mathcal{M}$ for all $j \geq 1$ because $\mathcal{M}$ is an algebra and thus is closed under set difference. Further, $\bigsqcup_{j \geq 1} B_j = \bigcup_{j \geq 1} A_j$.

We now show:

$$\mu^*\left(E \cap \bigsqcup_{j=1}^n B_j\right) = \sum_{j=1}^n \mu^*(E \cap B_j). \quad (3)$$

We show this by induction. The statement is trivial for $n = 1$. Suppose it is true for n . For $n + 1$, since $B_{j+1} \in \mathcal{M}$, we have:

$$\begin{aligned} \mu^*\left(E \cap \bigsqcup_{j=1}^{n+1} B_j\right) &= \mu^*\left(E \cap \bigsqcup_{j=1}^{n+1} B_j \cap B_{j+1}\right) + \mu^*\left(E \cap \bigsqcup_{j=1}^{n+1} B_j \cap B_{j+1}^c\right) \\ &= \mu^*(E \cap B_{j+1}) + \mu^*\left(E \cap \bigsqcup_{j=1}^n B_j\right) \\ &= \mu^*(E \cap B_{j+1}) + \sum_{j=1}^n \mu^*(E \cap B_j) \\ &= \sum_{j=1}^{n+1} \mu^*(E \cap B_j) \end{aligned}$$

Hence, $\mu^*\left(E \cap \bigcup_{j=1}^n A_j\right) = \mu^*\left(E \cap \bigsqcup_{j=1}^n B_j\right) = \sum_{j=1}^n \mu^*(E \cap B_j)$.
Using this in (2),

$$\mu^*(E) \geq \sum_{j=1}^n \mu^*(E \cap B_j) + \mu^*\left(E \setminus \bigcup_{j \geq 1} A_j\right)$$

This is true for all n . Hence,

$$\begin{aligned}
\mu^*(E) &\geq \sum_{j \geq 1} \mu^*(E \cap B_j) + \mu^*\left(E \setminus \bigcup_{j \geq 1} A_j\right) \\
&\geq \mu^*\left(\bigsqcup_{j \geq 1} (E \cap B_j)\right) + \mu^*\left(E \setminus \bigcup_{j \geq 1} A_j\right) \\
&\quad \text{(by subadditivity)} \\
&= \mu^*\left(E \cap \bigcup_{j \geq 1} A_j\right) + \mu^*\left(E \setminus \bigcup_{j \geq 1} A_j\right)
\end{aligned}$$

thereby showing that $\bigcup_{j \geq 1} A_j$ is in $\mathcal{M}$. Therefore, $\mathcal{M}$ is a σ -algebra.

2. We now show that $\mu^*|_{\mathcal{M}}$ is a complete measure.

We begin by showing that it is a measure. Since $\mathcal{M}$ is a σ -algebra, $\emptyset \in \mathcal{M}$, and $\mu^*(\emptyset) = 0$.

We further need to show countable additivity. Let $\{A_j\}_{j \geq 1}$ be a sequence of disjoint sets in $\mathcal{M}$. We want to show that $\mu^*\left(\bigsqcup_{j \geq 1} A_j\right) = \sum_{j \geq 1} \mu^*(A_j)$.

By substituting $E = \Omega$ in (3), we obtain finite additivity:

$$\mu^*\left(\bigsqcup_{j=1}^n A_j\right) = \sum_{j=1}^n \mu^*(A_j). \quad (4)$$

By sub-additivity,

$$\mu^*\left(\bigsqcup_{j \geq 1} A_j\right) \leq \sum_{j \geq 1} \mu^*(A_j). \quad (5)$$

Also,

$$\bigsqcup_{j \geq 1} A_j \supseteq \bigsqcup_{j=1}^n A_j$$

and so, by monotonicity, we obtain

$$\mu^*\left(\bigsqcup_{j \geq 1} A_j\right) \geq \mu^*\left(\bigsqcup_{j=1}^n A_j\right) = \sum_{j=1}^n \mu^*(A_j)$$

where the last equality is obtained from (4). Since this is true for all n , for $n \rightarrow \infty$, we get

$$\mu^*\left(\bigsqcup_{j \geq 1} A_j\right) \geq \sum_{j \geq 1} \mu^*(A_j). \quad (6)$$

From (5) and (6) we obtain the required equality. Thus, we have shown that $\mu^*|_{\mathcal{M}}$ is a measure. We now show that it is a complete measure.

Let $A \subseteq \Omega$ such that $\mu^*(A) = 0$. Then, for any $E \subseteq \Omega$, $E \cap A \subseteq A$ and so by monotonicity, $\mu^*(E \cap A) \leq \mu^*(A) \implies \mu^*(E \cap A) = 0$. Also, $E \supseteq E \cap A^c \implies \mu^*(E) \geq \mu^*(E \cap A^c)$. Thus,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

which implies that $A \in \mathcal{M}$. Therefore, all sets of measure zero are in $\mathcal{M}$, and so $\mu^*|_{\mathcal{M}}$ is a complete measure. $\square$

3 Pre-measure

Definition 3.1. (Pre-measure.) Let $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ be an algebra. Then, $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is said to be a *pre-measure* if:

1. $\mu_0(\emptyset) = 0$.
2. If $\{A_j\}_{j \geq 1}$ is a sequence of disjoint sets such that $\bigsqcup_{j \geq 1} A_j \in \mathcal{A}$, then $\mu_0\left(\bigsqcup_{j \geq 1} A_j\right) = \sum_{j \geq 1} \mu_0(A_j)$.

Proposition 3.2. Let Ω be a set and $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ be an algebra. Define $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ by

$$\mu^*(A) := \inf \left\{ \sum_{j \geq 1} \mu_0(A_j) : A \subseteq \bigcup_{j \geq 1} A_j \text{ and } A_j \in \mathcal{A} \forall j \geq 1 \right\}$$

for all $A \in \mathcal{P}(\Omega)$. Further, let $\mathcal{M}$ be the collection of μ^* -measurable sets. Then,

1. μ^* restricted to $\mathcal{A}$ is μ_0 .
2. $\mathcal{A} \subseteq \mathcal{M}$.

Proof. 1. Let $A \in \mathcal{A}$. We want to show that $\mu^*(A) = \mu_0(A)$. Set $A_1 = A$ and $A_j = \emptyset$ for all $j \geq 2$. Then, $\sum_{j \geq 1} \mu_0(A_j) = \mu_0(A)$ and so $\mu^*(A) \leq \mu_0(A)$. We now show the reverse inequality. Let $\{A_j\}_{j \geq 1}$ be an arbitrary covering of A with $A_j \in \mathcal{A}$ for all $j \geq 1$. Then, $A \subseteq \bigcup_{j \geq 1} A_j$. Define $B_1 = A_1$ and $B_j = A_j \setminus \bigcup_{k=1}^{j-1} A_k$. Then, $B_i \cap B_j = \emptyset$ for all $i \neq j$. Clearly, $B_j \in \mathcal{A}$ as $\mathcal{A}$ is an algebra and therefore is closed under finite unions and set difference. Also, $A \subseteq \bigsqcup_{j \geq 1} B_j = \bigcup_{j \geq 1} A_j$. As μ_0 is a pre-measure, $\mu_0(A) \leq \sum_{j \geq 1} \mu_0(B_j)$. But by definition, $B_j \subseteq A_j$ and therefore $\mu_0(B_j) \leq \mu_0(A_j)$, implying that $\sum_{j \geq 1} \mu_0(B_j) \leq \sum_{j \geq 1} \mu_0(A_j)$. Thus, $\mu_0(A) \leq \sum_{j \geq 1} \mu_0(A_j)$. Since the covering $\{A_j\}_{j \geq 1}$ was arbitrary, $\mu_0(A) \leq \mu^*(A)$. Hence, $\mu^*(A) = \mu_0(A)$.

2. Let $A \in \mathcal{A}$. We want to show that $A \in \mathcal{M}$, that is,

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for all $E \subseteq \Omega$.

Let $\varepsilon > 0$. There exists a covering $\{E_j\}_{j \geq 1}$ in $\mathcal{A}$ of $E \subseteq \Omega$ such that

$$\mu^*(E) \leq \sum_{j \geq 1} \mu_0(E_j) \leq \mu^*(E) + \varepsilon. \quad (7)$$

$E \subseteq \bigcup_{j \geq 1} E_j$. Let $F_1 = E_1$ and $F_j = E_j \setminus \bigcup_{k=1}^{j-1} E_k$. Then, $\{F_j\}_{j \geq 1}$ is a sequence of disjoint sets with $F_j \in \mathcal{A}$ for all $j \geq 1$, since $\mathcal{A}$ is an algebra and so is closed under finite unions and set difference. Also, $\bigsqcup_{j \geq 1} F_j = \bigcup_{j \geq 1} E_j$ and so $\{F_j\}_{j \geq 1}$ is also a covering of E .

Now,

$$E \cap A \subseteq \left(\bigsqcup_{j \geq 1} F_j \right) \cap A = \bigsqcup_{j \geq 1} (F_j \cap A).$$

Since $A, F_j \in \mathcal{A}$, $F_j \cap A \in \mathcal{A}$. Further, observe that μ^* is an outer measure by Proposition 1.2. Using sub-additivity of μ^* yields

$$\mu^*(E \cap A) \leq \mu^* \left(\bigsqcup_{j \geq 1} (F_j \cap A) \right) \leq \sum_{j \geq 1} \mu_0(F_j \cap A).$$

Similarly,

$$\mu^*(E \cap A^c) \leq \sum_{j \geq 1} \mu_0(F_j \cap A^c).$$

Therefore,

$$\begin{aligned} \mu^*(E \cap A) + \mu^*(E \cap A^c) &\leq \sum_{j \geq 1} \mu_0(F_j \cap A) + \sum_{j \geq 1} \mu_0(F_j \cap A^c) \\ &= \sum_{j \geq 1} (\mu_0(F_j \cap A) + \mu_0(F_j \cap A^c)) \\ &= \sum_{j \geq 1} \mu_0(F_j). \end{aligned}$$

Since $F_j \subseteq E_j$, $\mu_0(F_j) \leq \mu_0(E_j)$. Hence, $\sum_{j \geq 1} \mu_0(F_j) \leq \sum_{j \geq 1} \mu_0(E_j)$, which gives

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \sum_{j \geq 1} \mu_0(E_j). \quad (8)$$

(7) and (8) imply

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E) + \varepsilon$$

but since this is true for all $\varepsilon > 0$, we obtain $\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E)$, as required. $\square$

4 Extension of Pre-measure to Measure

Theorem 4.1. Let $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ be an algebra and let $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ be a pre-measure. Define $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ by

$$\mu^*(A) := \inf \left\{ \sum_{j \geq 1} \mu_0(A_j) : A \subseteq \bigcup_{j \geq 1} A_j \text{ and } A_j \in \mathcal{A} \ \forall j \geq 1 \right\}$$

for all $A \in \mathcal{P}(\Omega)$. Further, let $\mathcal{F}(\mathcal{A})$ be the σ -algebra generated by $\mathcal{A}$ and μ be the restriction of μ^* to $\mathcal{F}(\mathcal{A})$. Then:

1. μ is a measure which extends the pre-measure μ_0 to $\mathcal{F}(\mathcal{A})$.
2. Let ν be any other measure which extends μ_0 to $\mathcal{F}(\mathcal{A})$. Then, $\nu(E) \leq \mu(E)$ for all $E \in \mathcal{F}(\mathcal{A})$, and if $\mu(E) < \infty$, then $\nu(E) = \mu(E)$. Further, if μ_0 is σ -finite, then μ is the unique extension of μ_0 to $\mathcal{F}(\mathcal{A})$.

Proof. 1. We want to show that $\mu|_{\mathcal{A}} = \mu_0$. Let $\mathcal{M}$ be the collection of μ^* -measurable sets. By Theorem 2.2, $\mathcal{M}$ is a σ -algebra, and by Proposition 3.2, $\mathcal{A} \subseteq \mathcal{M}$. Therefore, $\mathcal{F}(\mathcal{A}) \subseteq \mathcal{M}$. By Proposition 3.2, we also have that $\mu^*|_{\mathcal{A}} = \mu_0$. Hence, $\mu|_{\mathcal{A}} = \mu_0$. Further, $\mu^*|_{\mathcal{M}}$ is a measure, as we showed in Theorem 2.2. This implies that $\mu^*|_{\mathcal{M}}$ possesses the property of countable additivity, and thus, $\mu^*|_{\mathcal{F}(\mathcal{A})}$ also possesses it, that is, μ is a measure.

2. We first show that if ν is any other measure which extends μ_0 to $\mathcal{F}(\mathcal{A})$, then $\nu(E) \leq \mu(E)$ for all $E \in \mathcal{F}(\mathcal{A})$. Let $\{E_j\}_{j \geq 1}$ be a covering of E in $\mathcal{A}$. Then, by the sub-additivity of ν , $\nu(E) \leq \sum_{j \geq 1} \nu(E_j)$. But since ν is an extension of μ_0 and $E_j \in \mathcal{A}$, $\nu(E_j) = \mu_0(E_j)$ and therefore, $\nu(E) \leq \sum_{j \geq 1} \mu_0(E_j)$. This is true for all coverings of E in $\mathcal{A}$ and therefore, $\nu(E) \leq \mu^*(E)$ by definition of μ^* . Now, $E \in \mathcal{F}(\mathcal{A})$ and $\mu = \mu^*|_{\mathcal{F}(\mathcal{A})}$ and thus, $\nu(E) \leq \mu(E) = \mu^*(E)$.

Let $\mu(E) < \infty$. Suppose $E \subseteq \bigcup_{j \geq 1} E_j$ with $E_j \in \mathcal{A}$ for all $j \geq 1$. As usual, let $\{F_j\}_{j \geq 1}$ be the sequence of disjoint sets in $\mathcal{A}$ such that

$\bigsqcup_{j=1}^n F_j = \bigcup_{j=1}^n E_j$. Then,

$$\begin{aligned}
\nu \left(\bigcup_{j \geq 1} E_j \right) &= \lim_{n \rightarrow \infty} \nu \left(\bigcup_{j=1}^n E_j \right) = \lim_{n \rightarrow \infty} \nu \left(\bigsqcup_{j=1}^n F_j \right) \\
&= \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu_0(F_j) \\
&= \lim_{n \rightarrow \infty} \mu \left(\bigsqcup_{j=1}^n F_j \right) \quad (\text{as } \mu \text{ is also an extension of } \mu_0) \\
&= \lim_{n \rightarrow \infty} \mu \left(\bigcup_{j=1}^n E_j \right)
\end{aligned}$$

Hence,

$$\nu \left(\bigcup_{j \geq 1} E_j \right) = \mu \left(\bigcup_{j \geq 1} E_j \right). \quad (9)$$

As μ is a restriction of μ^* and by the way μ^* was defined, there exists a covering $\{E_j\}_{j \geq 1}$ of E in $\mathcal{A}$ such that $\mu(E) \leq \sum_{j \geq 1} \mu_0(E_j) \leq \mu(E) + \varepsilon$. Then, $\{F_j\}_{j \geq 1}$ is also a covering of E such that F_j are disjoint and $F_j \in \mathcal{A}$, with $\bigsqcup_{j \geq 1} F_j = \bigcup_{j \geq 1} E_j$. Since $F_j \subseteq E_j$, we have that $\mu_0(F_j) \leq \mu_0(E_j)$ and so,

$$\mu \left(\bigsqcup_{j \geq 1} F_j \right) = \sum_{j \geq 1} \mu(F_j) = \sum_{j \geq 1} \mu_0(F_j) \leq \sum_{j \geq 1} \mu_0(E_j) \leq \mu(E) + \varepsilon.$$

That is, we just showed that for any $E \in \mathcal{F}(\mathcal{A})$, there exists a covering $\{F_j\}_{j \geq 1}$ in $\mathcal{A}$ such that if $F = \bigsqcup_{j \geq 1} F_j$, then $\mu(F) \leq \mu(E) + \varepsilon$. Also, since $E \subseteq F$, $\mu(F) = \mu(E) + \mu(F \setminus E)$, and so $\mu(F \setminus E) = \mu(F) - \mu(E) \leq \varepsilon$ (here we used the fact that $\mu(E) < \infty$). Now,

$$\begin{aligned}
\mu(E) &\leq \mu \left(\bigcup_{j \geq 1} E_j \right) = \nu \left(\bigcup_{j \geq 1} E_j \right) && (\text{from (9)}) \\
&= \nu(F) \\
&= \nu(E) + \nu(F \setminus E) && (\text{as } \nu \text{ is a measure}) \\
&\leq \nu(E) + \mu(F \setminus E) \\
&\leq \nu(E) + \varepsilon.
\end{aligned}$$

This is true for all $\varepsilon > 0$ and thus $\mu(E) \leq \nu(E)$. Combining this with the fact that $\nu(E) \leq \mu(E)$ yields the equality $\nu(E) = \mu(E)$ if $\mu(E) < \infty$.

Finally, suppose that μ_0 is σ -finite. That is, $\Omega = \bigcup_{j \geq 1} E_j$ with $E_j \in \mathcal{A}$ and $\mu_0(E_j) < \infty$ for all $j \geq 1$. Then, for $F_1 = E_1$ and $F_j = E_j \setminus \bigcup_{k=1}^{j-1} E_k$

for $j \geq 2$, $\Omega = \bigsqcup_{j \geq 1} F_j$, where F_j are disjoint and belong to $\mathcal{A}$. Also, note that since $F_j \subseteq E_j$, $\mu_0(F_j) < \infty$. Then, if ν is any other extension of μ_0 to $\mathcal{F}(\mathcal{A})$, for any $A \in \mathcal{F}(\mathcal{A})$,

$$\nu(A) = \nu(A \cap \Omega) = \nu\left(A \cap \bigsqcup_{j \geq 1} F_j\right) = \sum_{j \geq 1} \nu(A \cap F_j)$$

and since $A \cap F_j \subseteq F_j$, $\mu(A \cap F_j) < \infty$, implying that $\nu(A \cap F_j) = \mu(A \cap F_j)$, and thus,

$$\sum_{j \geq 1} \nu(A \cap F_j) = \sum_{j \geq 1} \mu(A \cap F_j) = \mu\left(A \cap \bigsqcup_{j \geq 1} F_j\right) = \mu(A).$$

Therefore, $\nu(A) = \mu(A)$, showing that μ is indeed the unique extension of μ_0 to $\mathcal{F}(\mathcal{A})$. □